

Module 2

Relational Model and E-R Modeling

Topic

Mapping ER Model to Relational Model

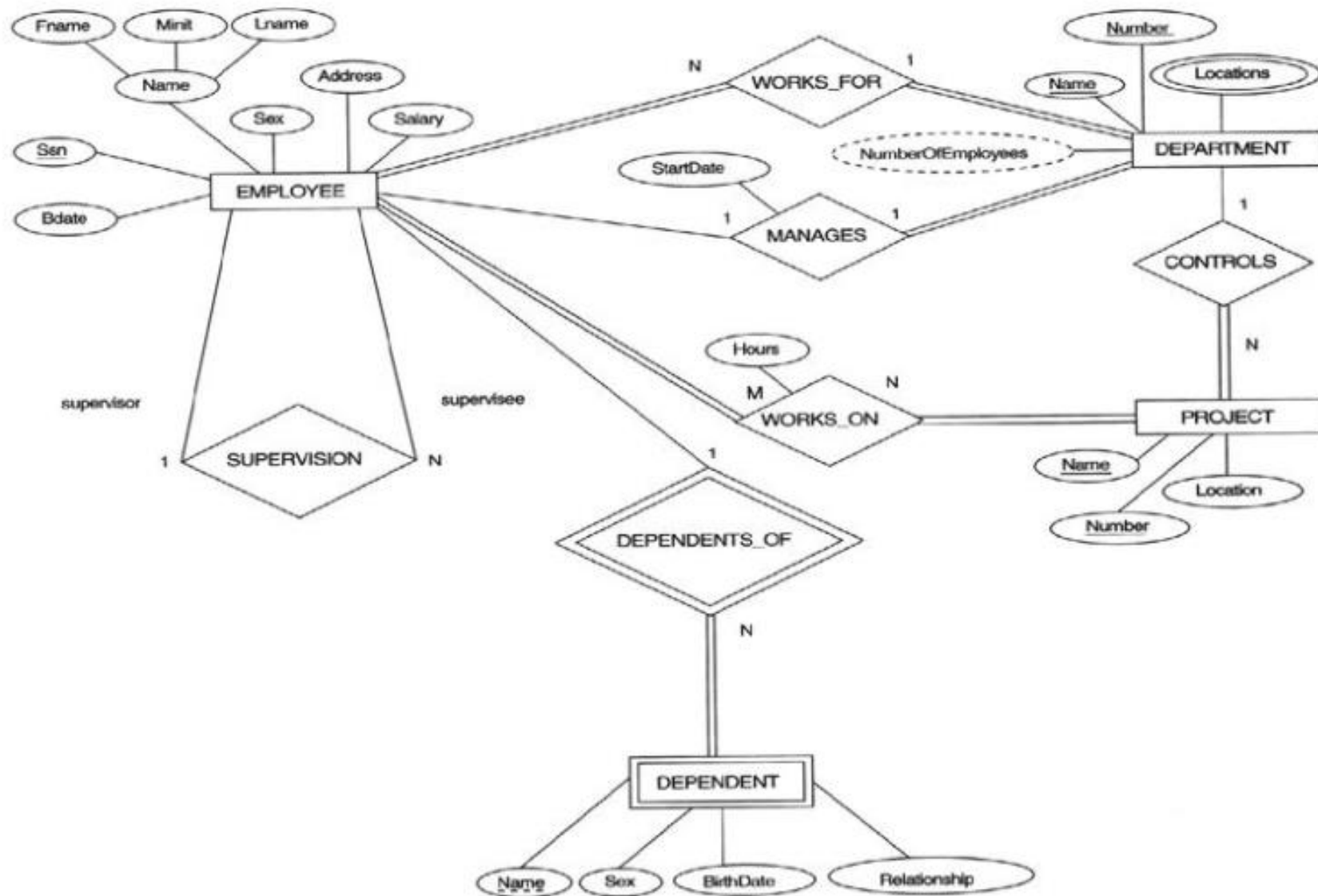
(Abraham Silberschatz)

ER to Relational Mapping – Algorithm

7-Step Process:

1. Map **Regular Entity** Types
2. Map **Weak Entity** Types
3. Map **Binary 1:1 Relation** Types
4. Map **Binary 1:N Relationship** Types.
5. Map **Binary M:N Relationship** Types.
6. Map **Multivalued attributes**.
7. Map **N-ary Relationship** Types.

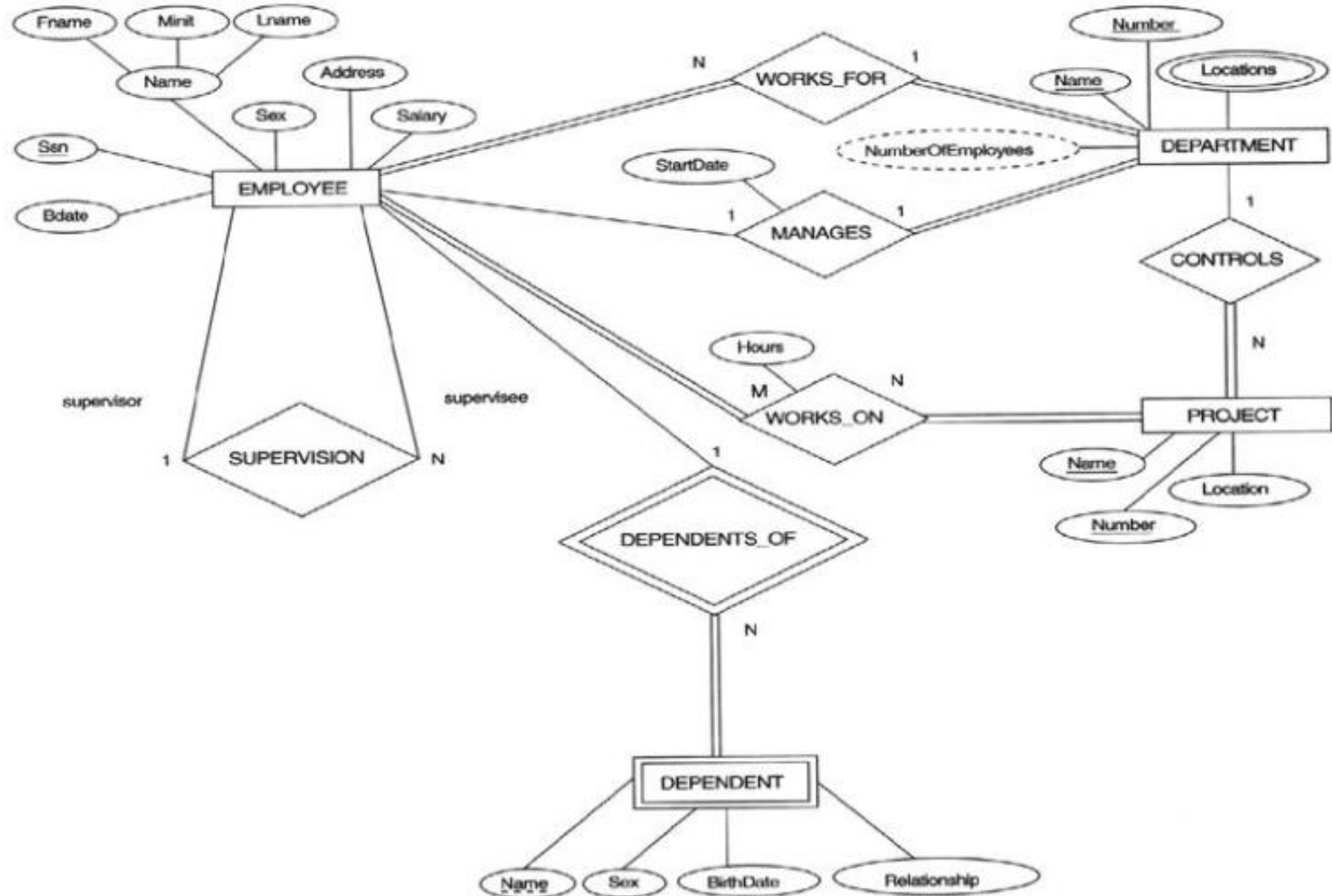
The ERD for the COMPANY database



Step 1: Mapping of Regular Entity Types

- For each regular (strong) entity type E in the ER schema, create a relation R that includes all the simple attributes of E .
- Choose one of the key attributes of E as the primary key for R .
- If the chosen key of E is composite, the set of simple attributes that form it will together form the primary key of R .

The ERD for the COMPANY database

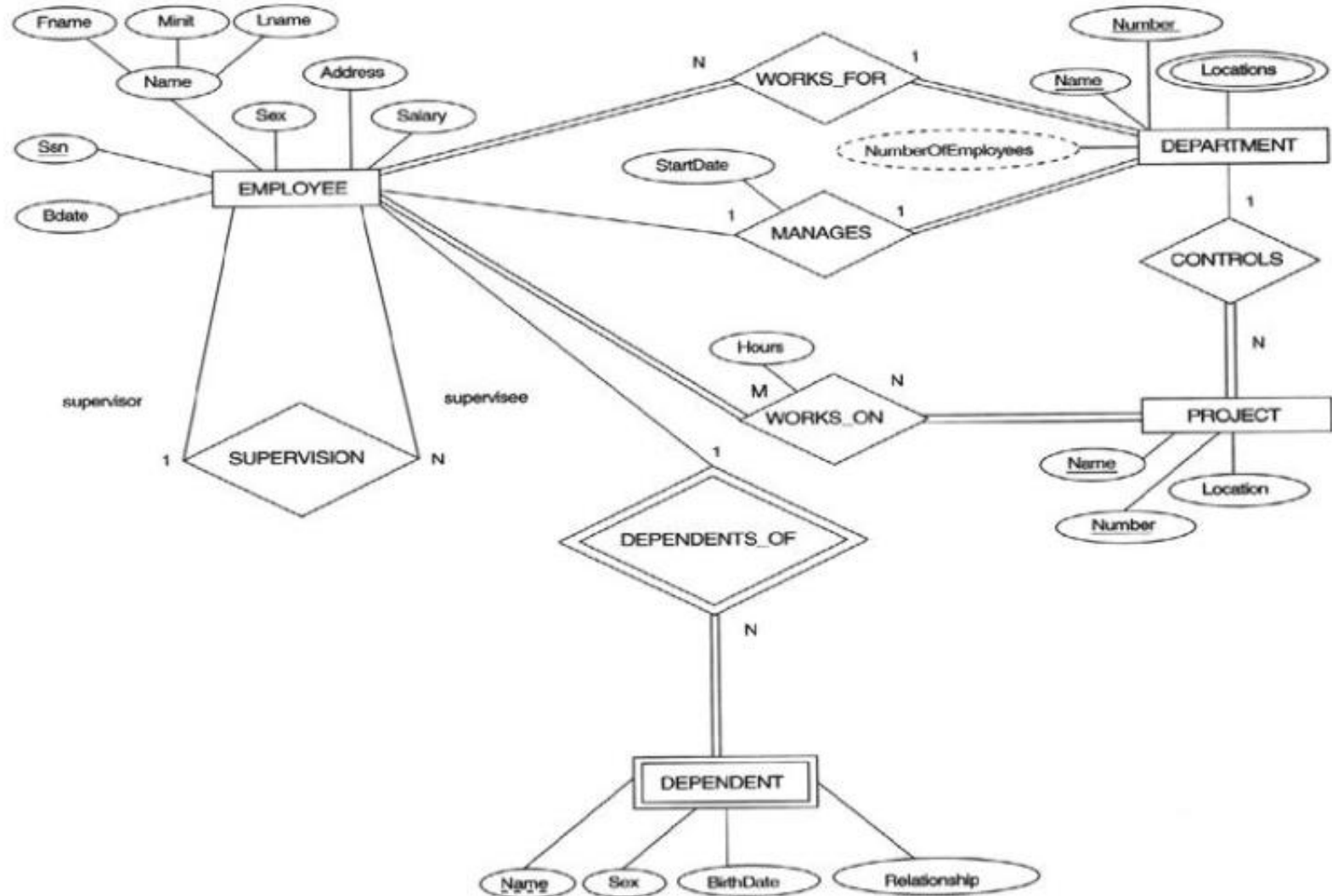


EMPLOYEE							
Fname	Minit	Lname	Sex	Address	Salary	<u>Ssn</u>	Bdate
DEPARTMENT							
Name	<u>Number</u>						
PROJECT							
Name	<u>Number</u>	Location					

Step 2: Mapping of Weak Entity Types

- For each weak entity type W in the ER schema with owner entity type E, create a relation R & include all simple attributes (or simple components of composite attributes) of W as attributes of R.
- Also, include as foreign key attributes of R the primary key attribute(s) of the relation(s) that correspond to the owner entity type(s).
- The primary key of R is the *combination of* the primary key(s) of the owner(s) and the partial key of the weak entity type W, if any.

The ERD for the COMPANY database



EMPLOYEE

Fname	Minit	Lname	Sex	Address	Salary	<u>Ssn</u>	Bdate
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DEPARTMENT

Name	<u>Number</u>
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DEPENDENT

<u>ESSN</u>	<u>Name</u>	Sex	Bdate	Relationship
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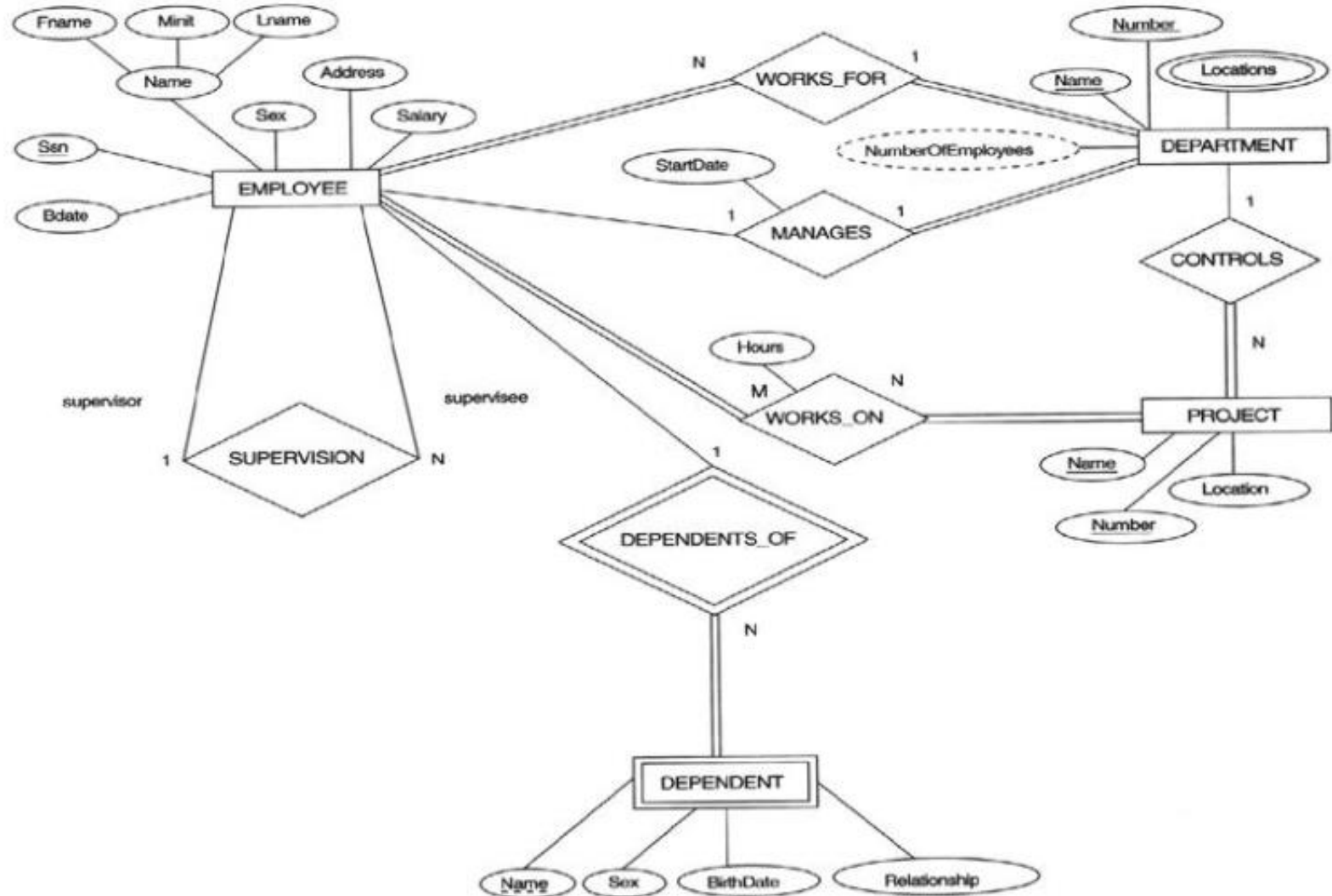
PROJECT

Name	<u>Number</u>	Location
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Step 3: Mapping of Binary 1:1 Relation Types

- For each binary 1:1 relationship type R in the ER schema, identify the relations S and T that correspond to the entity types participating in R .
- Choose one of the relations—say S —and include a foreign key in S the primary key of T . It is better to choose an entity type with **total participation** in R in the role of S .

The ERD for the COMPANY database



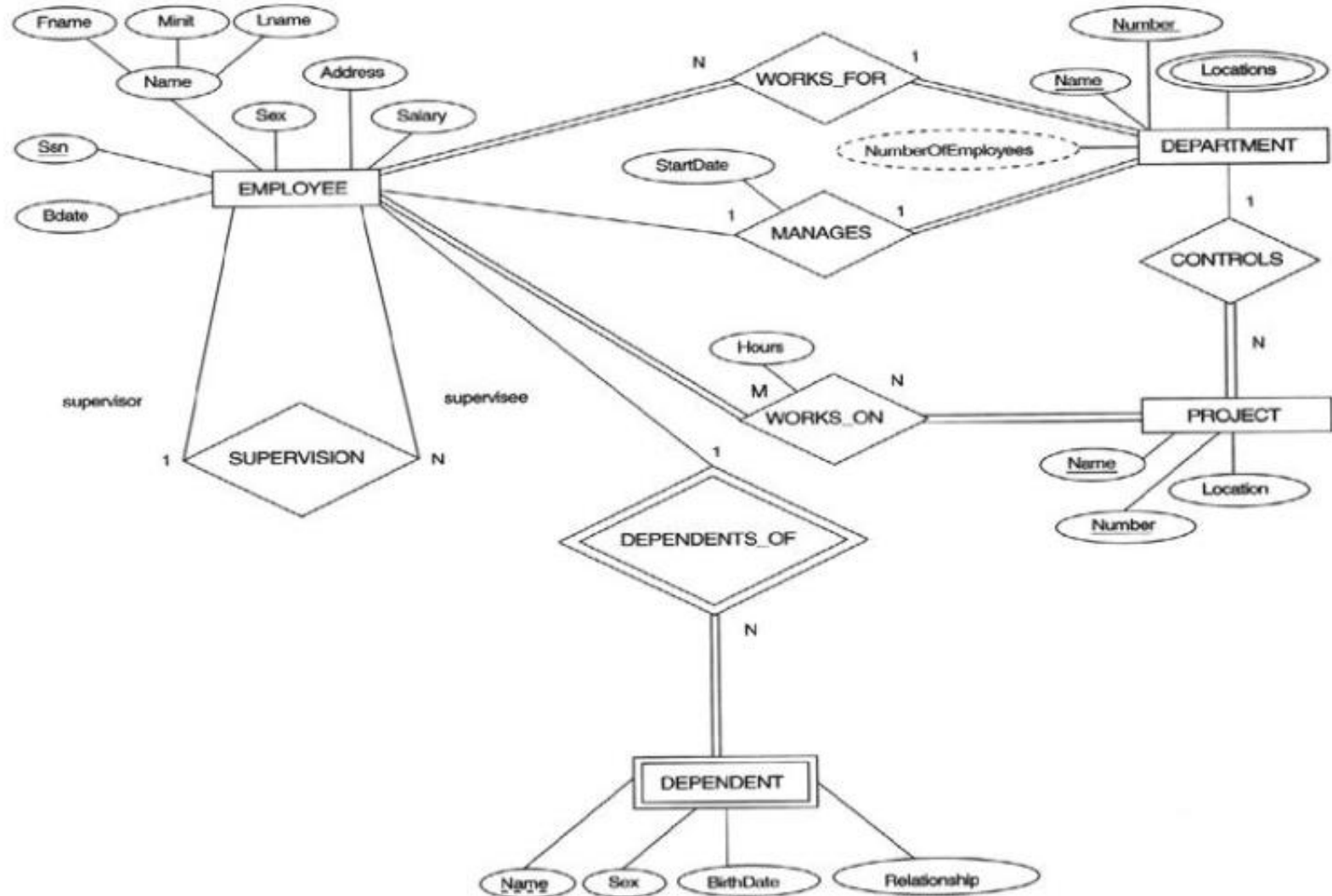
EMPLOYEE							
Fname	Minit	Lname	Sex	Address	Salary	<u>Ssn</u>	Bdate
DEPARTMENT							
Name	<u>Number</u>	Mgr_Ssn	Mgr_StartDate				
DEPENDENT							
<u>ESSN</u>	<u>Name</u>	Sex	Bdate	Relationship			
PROJECT							
Name	<u>Number</u>	Location					

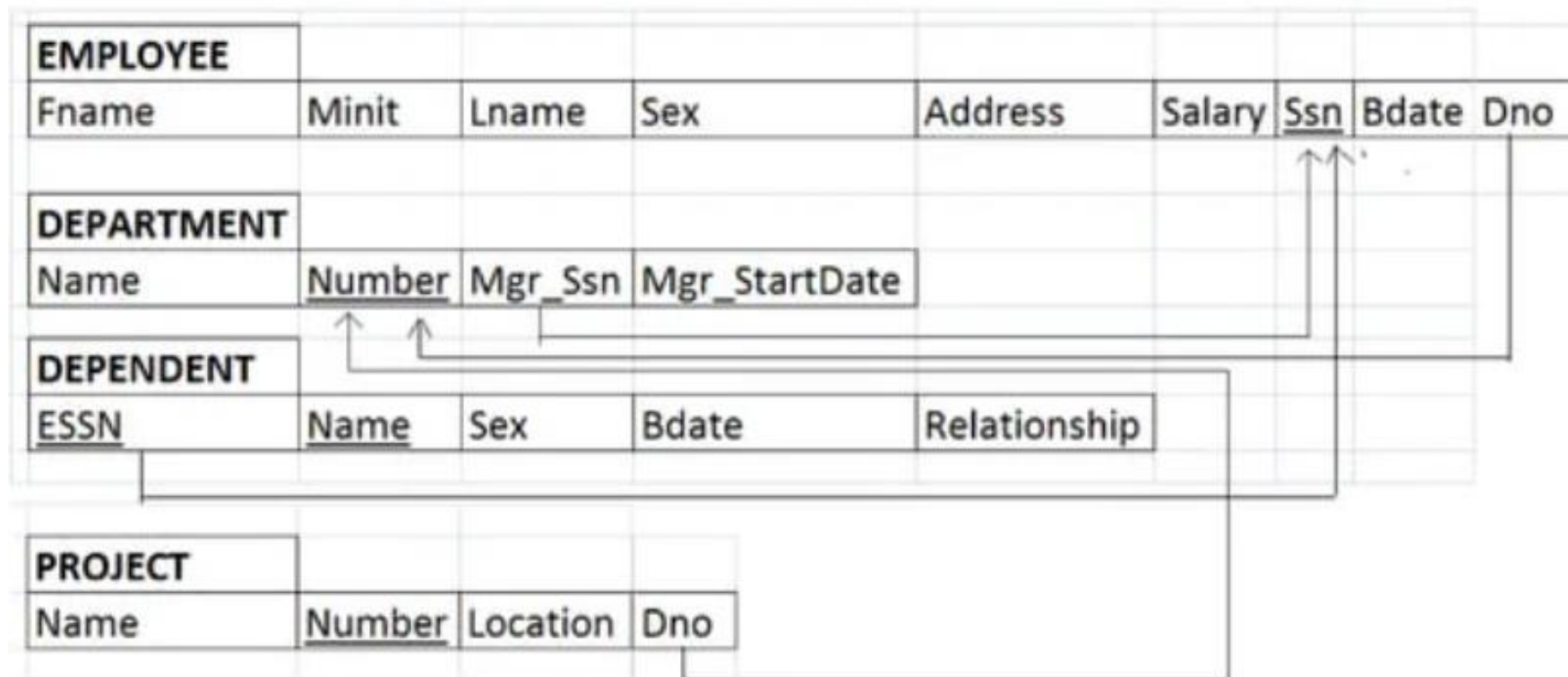
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graph TD
    DEPT[DEPARTMENT] -- "Mgr_Ssn" --> EMP[EMPLOYEE]
    DEPT -- "Mgr_StartDate" --> EMP
    DEPEND[DEPENDENT] -- "ESSN" --> EMP
    DEPEND -- "Name" --> EMP
    PROJECT[PROJECT]
```

Step 4: Mapping of Binary 1:N Relationship Types

- For each regular binary 1:N relationship type R , identify the relation S that represent the participating entity type at the N-side of the relationship type.
- Include as foreign key in S the primary key of the relation T that represents the other entity type participating in R .
- Include any simple attributes of the 1:N relation type as attributes of S .

The ERD for the COMPANY database

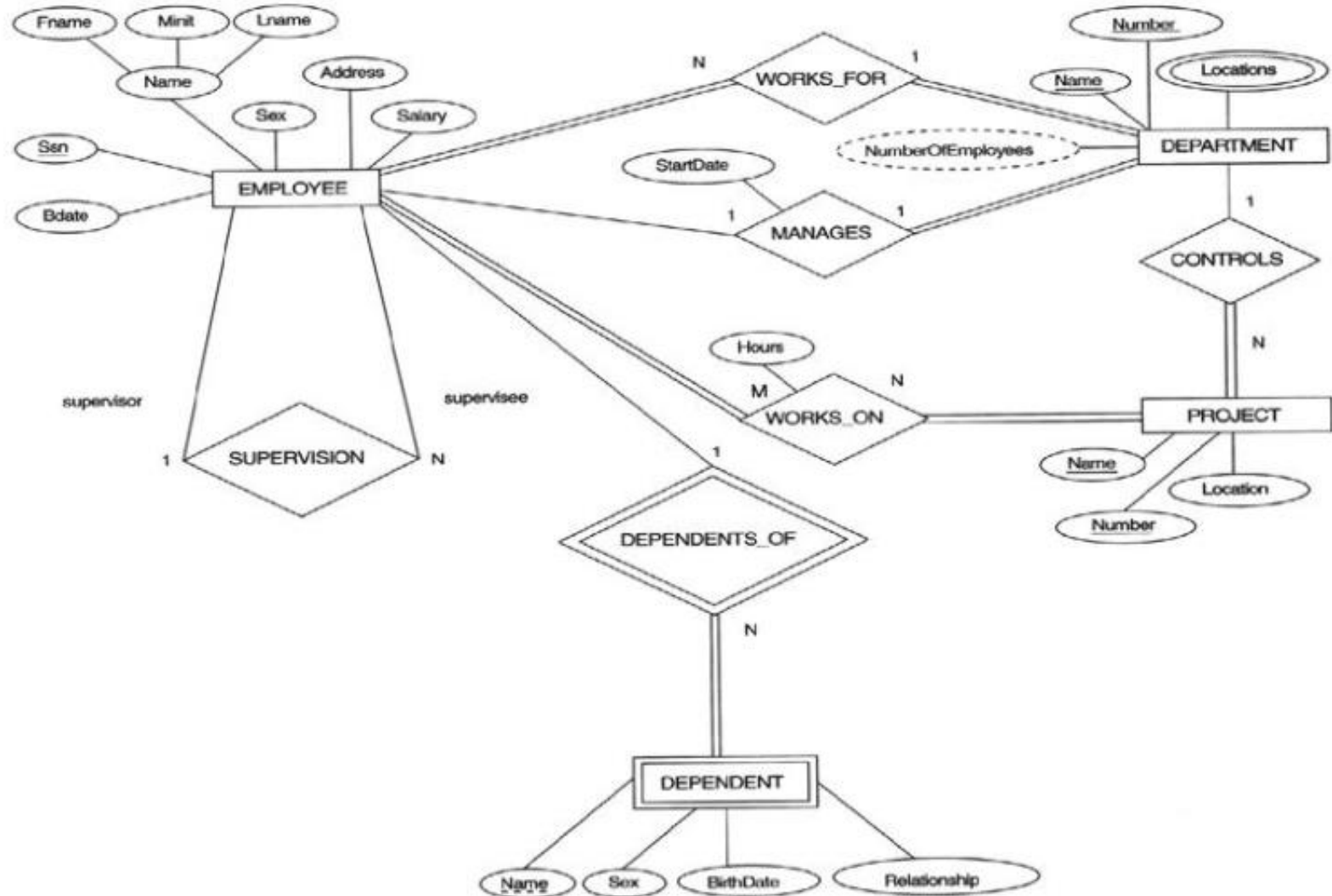




Step 5: Mapping of Binary M:N Relationship Types

- For each regular binary M:N relationship type R, *create a new relation S* to represent R.
- Include as foreign key attributes in S the primary keys of the relations that represent the participating entity types; *their combination will form the primary key* of S.
- Also include any simple attributes of the M:N relationship type (or simple components of composite attributes) as attributes of S.

The ERD for the COMPANY database

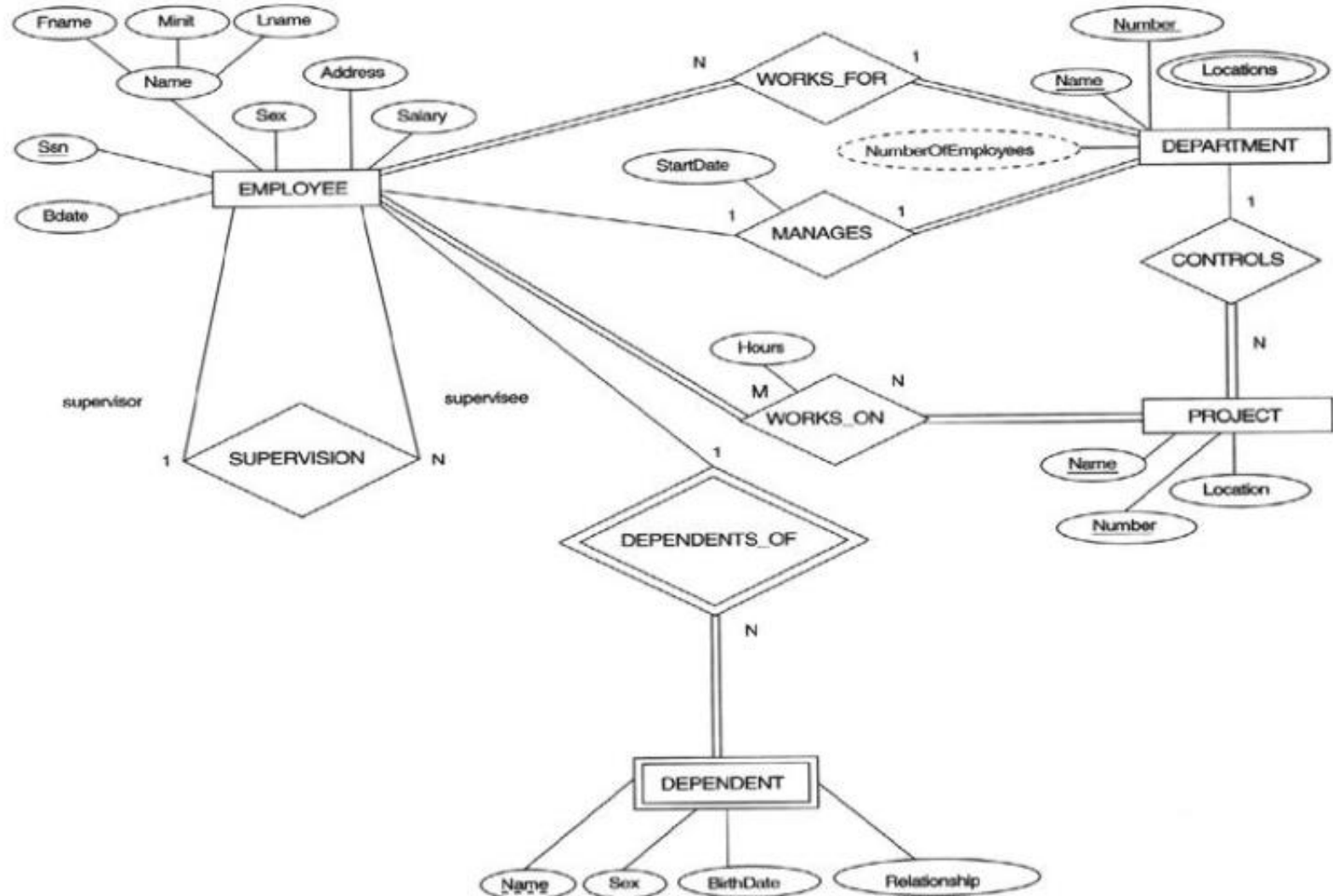


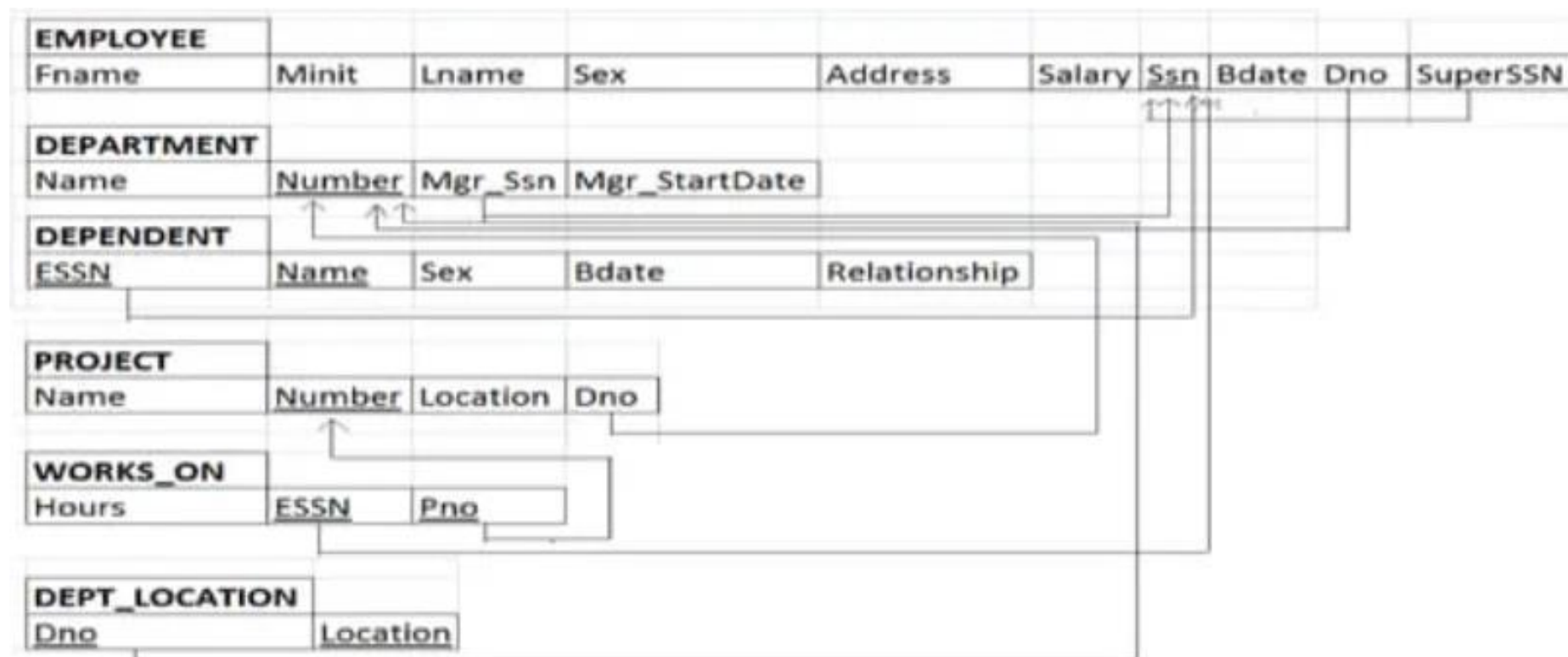


Step 6: Mapping of Multivalued attributes

- For each multivalued attribute A, create a new relation R.
- This relation R will include an attribute corresponding to A, plus the primary key attribute K—as a foreign key in R—of the relation that represents the entity type of relationship type that has A as an attribute.
- The primary key of R is the combination of A and K. If the multivalued attribute is composite, we include its simple components.

The ERD for the COMPANY database

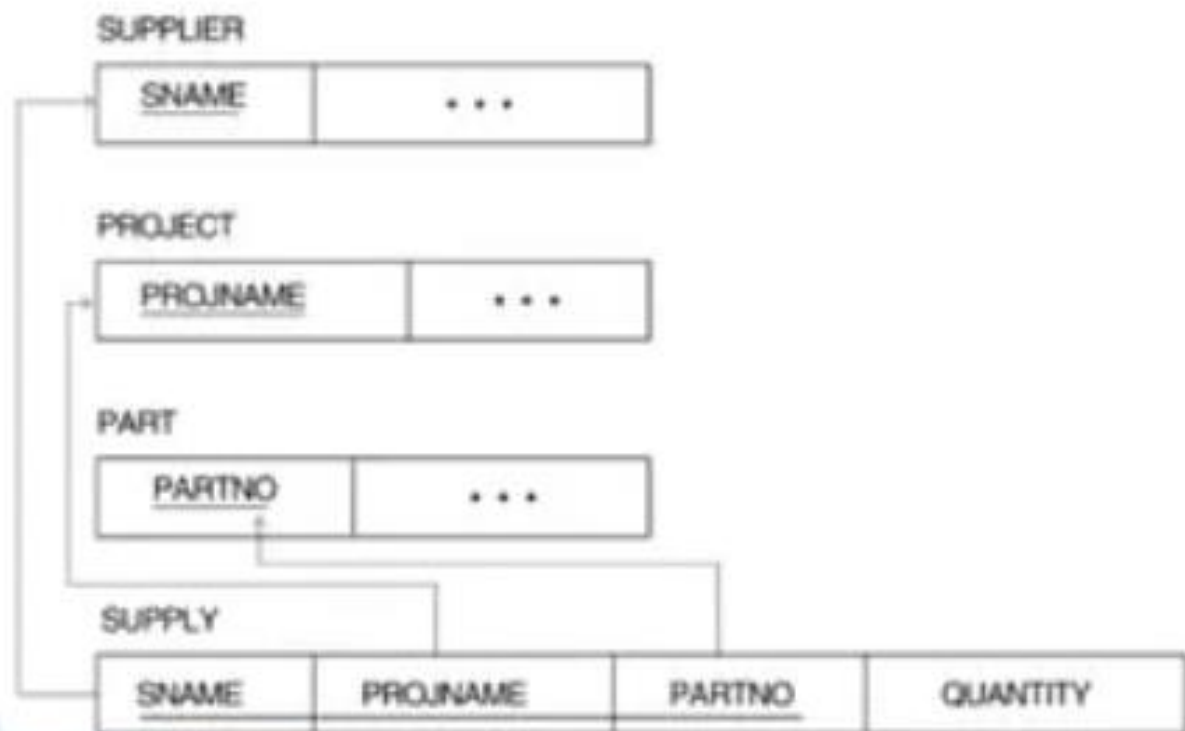
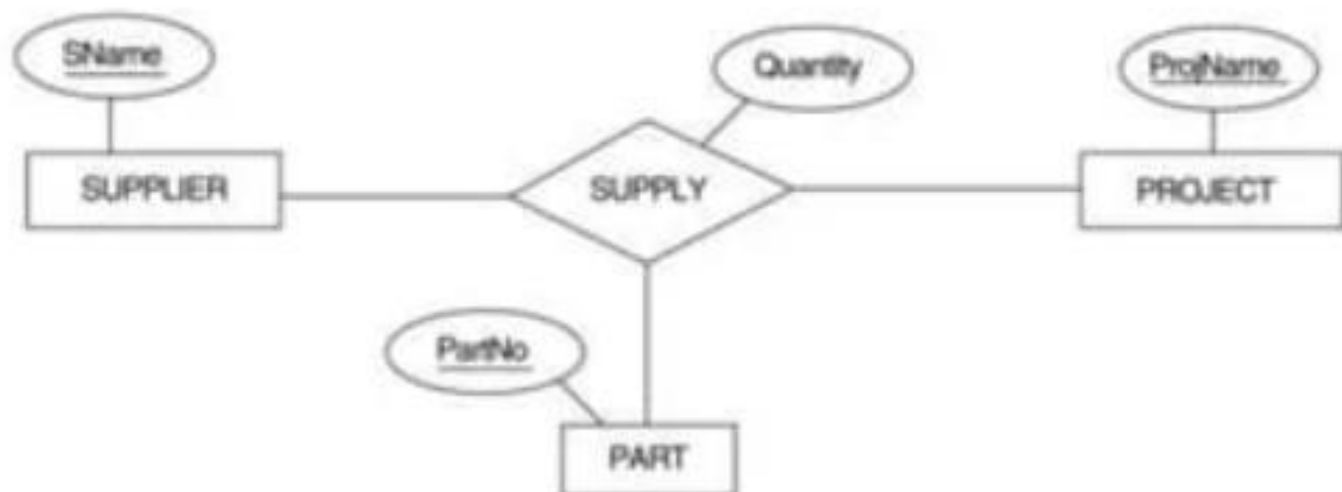




Step 7: Mapping of N-ary Relationship Types

- For each n -ary relationship type R , where $n > 2$, create a new relationship S to represent R .
- Include as foreign key attributes in S the primary keys of the relations that represent the participating entity types.
- Also include any simple attributes of the n -ary relationship type (or simple components of composite attributes) as attributes of S .

(a)



Transform ERD to Relational Schema

